F7AB - Actuarial and Financial Mathematics

Tutorial 4 – Force of interest and continuously-paid cashflows

Question 1 An investment of £1000 has been made at time 0. The force of interest at time t is $\delta(t) = 0.05 + 0.0001t$, for all $t > 0$.

1. Calculate the accumulation of the investment at time 10.
2. Calculate the accumulation of the investment at time 20.
3. Calculate the constant, effective annual rate of interest that will give the same accumulation as in part 2 above.

Question 2 Suppose that $A(0) = 1$ and that the force of interest is

$$\delta(t) = \alpha + \frac{\beta}{1+\beta t} \quad \text{for all } t > 0, \text{ for constants } \alpha, \beta > 0.$$

1. Find an expression for the accumulation function $A(t)$ for $t > 0$.
2. For $\beta = 1$ and $\alpha = 0.05$, calculate:
   (a) the accumulation at time 5 of £100 paid at time 0.
   (b) the present value at time 0 of £200 paid at time 10.
   (c) the value at time 5 of £500 paid at time 11.

Question 3 The force of interest is $\delta(t) = 0.04 - 0.002t + 0.0003t^2$, for all $t > 0$.

1. Calculate the accumulation up to time 10 of £500 invested at time 0, plus a further £750 invested at time 5.
2. Calculate the constant, effective annual rate of interest that will give the same accumulation as in part 1 above.

Question 4 The force of interest is given by

$$\delta(t) = \begin{cases} 0.04 & \text{for } 0 \le t < 5 \\ 0.05 & \text{for } 5 \le t < 8 \\ 0.055 & \text{for } 8 \le t \end{cases}$$

1. Derive expressions for the accumulation function $A(t)$ given $A(0) = 1$.
2. Calculate the present value at time 0 of £700 due at time 10.
3. Calculate the value at time 6 of £1 000 due at time 15.

Question 5 The force of interest is given by

$$\delta(t) = 0.1 - 0.05e^{-0.1t}, \quad \text{for all } t > 0.$$

Find the value at time 2 of £10 000 payable at time 8.

Question 6 Starting from $t = 2$ years, a continuous payment stream is received for 5 years and the rate of payment is £500 p.a. In addition, a lump sum payment of £800 is received at $t = 1.5$ year. The annual force of interest is

$$\delta(t) = \begin{cases} 5\% & \text{for } 0 \leq t < 2 \\ 7\% & \text{for } 2 \leq t < 10. \end{cases}$$

Calculate

1. the present value at time 0 of the payments.

2. the accumulated value at time 10 of the payments.

Question 7 On the basis of an effective annual interest rate of 8%, find the values of: (i) $\bar{a}_{\overline{5}|}$ (ii) $\bar{a}_{\overline{9}|}$ (iii) ${}_{5|}\bar{a}_{\overline{4}|}$ (iv) $\bar{s}_{\overline{9}|}$ (v) $(I\bar{a})_{\overline{15}|}$ (vi) $(\bar{I}\bar{a})_{\overline{15}|}$.

Question 8 Let $(\bar{I}\bar{a})_{\overline{n}|}$ denote the present value at time 0 on the basis of a constant force of interest δ p.a. of a continuously increasing annuity which is payable continuously at a rate $\rho(t) = t$ p.a. for n years.

1. Show that
$$(\bar{I}\bar{a})_{\overline{n}|} = \frac{1}{\delta}\left(\bar{a}_{\overline{n}|} - nv^n\right).$$

2. Without doing any further calculations, explain why
$$(\bar{I}\bar{a})_{\overline{n}|} < (I\bar{a})_{\overline{n}|}.$$

Question 9 The force of interest is constant and equal to $\delta = 0.04$. Payments are received continuously for 15 years at the rate

$$\rho(t) = \begin{cases} 1000 & \text{for } 0 \leq t < 10 \\ 1500 & \text{for } 10 \leq t < 15. \end{cases}$$

Calculate the present value at time 0 of these cashflows.

Question 10 Payments are made continuously into an account such that the rate of payment at time t years is £$(100 - 5t)$ per annum. The account accumulates continuously in such a way that the force of interest at time t years is $\delta(t) = 0.1 - 0.005t$. Given that payments start at time $t = 0$, find the amount of the account after 20 years.

Question 11 The force of interest, $\delta(t)$, is a function of time and at any time t, measured in years, is given by the formula

$$\delta(t) = \begin{cases} 0.05 + 0.001t & \text{for } 0 \leq t \leq 20 \\ 0.05 & \text{for } t > 20. \end{cases}$$

1. Derive and simplify as far as possible expressions for $V(0,t)$, where $V(0,t)$ is the present value at time 0 of a unit sum of money paid at time t.

2. (a) Calculate the present value at time $t = 0$ of £100 due at the end of 25 years.

   (b) Calculate the rate of discount per annum convertible quarterly implied by the transaction in part (a).

3. A continuous payment stream is received at rate $30e^{-0.015t}$ units per annum between $t = 20$ and $t = 25$. Calculate the accumulated value of the payment stream at time $t = 25$.

Answers: Q1.1. 1656.99; Q1.2. 2773.19; Q1.3. 5.232% p.a.; Q2.2.a. £770.42; Q2.2.b. £11.03; Q2.2.c. £185.20; Q3.1. 1673.49; Q3.2. 4.1804% per time unit; Q4.2. 441.90; Q4.3. 615.70; Q5. 6601.43; Q6.1. 2650.83; Q6.2. 5128.81; Q7.i. 4.15037; Q7.ii. 6.49356; Q7.iii. 2.34319; Q7.iv. 12.9807; Q7.v. 58.6740; Q7.vi. 54.1682; Q9. 12798.57; Q10. £1718.28; Q11.2.a. £23.46; Q11.2.b. 0.05758; Q11.3. 121.82.

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Solutions to Tutorial 4 – Force of interest and continuously-paid cashflows

Solution 1 Let $C(t)$ = accumulated amount at time t.

$$\begin{aligned} C(t) &= 1000 \exp\left\{\int_0^t \delta(s)ds\right\} \\ &= 1000 \exp\left\{\int_0^t (0.05 + 0.0001s)ds\right\} \\ &= 1000 \exp\left\{0.05t + 0.00005t^2\right\} \end{aligned}$$

1. $C(10) = 1000e^{0.505} = 1656.99$
2. $C(20) = 1000e^{1.02} = 2773.19$
3. The constant, effective annual rate i satisfies

$$(1+i)^{20} = \frac{C(20)}{C(0)} = \frac{2773.19}{1000} \qquad \Rightarrow \qquad 1+i = \left(\frac{2773.19}{1000}\right)^{1/20} \qquad \Rightarrow \qquad i = 5.232\%.$$

Solution 2

1. $$\int_0^t \delta(s)ds = \int_0^t \left(\alpha + \frac{\beta}{1+\beta s}\right) ds = \alpha t + [\log(1+\beta s)]_{s=0}^{s=t} = \alpha t + \log(1+\beta t).$$

   Hence

$$A(t) = A(0) \exp\left\{\int_0^t \left(\alpha + \frac{\beta}{1+\beta s}\right) ds\right\} = \exp\left\{\alpha t + \log(1+\beta t)\right\} = (1+\beta t)e^{\alpha t}.$$

2. With $\beta = 1$ and $\alpha = 0.05$, the accumulation function at time t is $A(t) = (1+t)e^{0.05t}$.

   (a) $100A(5) = 600e^{0.05\times 5} = £770.42$.

   (b)
$$\frac{200}{A(10)} = \frac{200}{11e^{0.05\times 10}} = £11.03.$$

   (c)
$$500\frac{A(5)}{A(11)} = 500\frac{6e^{0.05\times 5}}{12e^{0.05\times 11}} = £185.20.$$

Solution 3

1.
$$\begin{aligned} A(t,T) &= \exp\left\{\int_t^T \delta(s)ds\right\} \\ &= \exp\left\{\int_t^T (0.04 - 0.002s + 0.0003s^2)ds\right\} \\ &= \exp\left\{0.04(T-t) - 0.001(T^2-t^2) + 0.0001(T^3-t^3)\right\} \end{aligned}$$

Hence

$$\begin{aligned} AV(10) &= 500A(0,10) + 750A(5,10) \\ &= 500\exp\left\{0.04(10-0) - 0.001(10^2-0^2) + 0.0001(10^3-0^3)\right\} \\ &\quad + 750\exp\left\{0.04(10-5) - 0.001(10^2-5^2) + 0.0001(10^3-5^3)\right\} \\ &= 500\exp\{0.4\} + 750\exp\{0.2125\} \\ &= 745.912 + 927.575 \\ &= 1673.49 \end{aligned}$$

2. The constant, effective annual rate i is the solution to the equation of value:

$$500 + 750v^5 = 1673.49v^{10}$$

Solving this as a quadratic equation in $x = v^5$ we find the two solutions $x = 0.814836$ and $x = -0.366671$. However, as v^5 must be positive, the only admissible solution is $v^5 = 0.814836$. Hence, the yield is $i = 0.041804$.

Solution 4

1.

$$A(t) = \begin{cases} e^{0.04t} & \text{for } 0 < t < 5 \\ e^{0.2+0.05(t-5)} & \text{for } 5 \le t < 8 \\ e^{0.35+0.055(t-8)} & \text{for } 8 \le t \end{cases}$$

The discount function is $V(t,T) = A(t)/A(T)$.

2. Hence the PV at time 0 of 700 due at time 10 is

$$700\,A(0)/A(10) = 700\,e^{-0.35-0.055\times 2} = 700\,e^{-0.46} = 441.90$$

3. The PV at time 6 of 1 000 due at time 15 is

$$1\,000\,A(6)/A(15) = 1\,000\,e^{0.2+0.05}e^{-0.35-0.055\times 7} = 1\,000\,e^{0.25}e^{-0.735} = 1\,000\,e^{-0.485} = 615.70$$

Solution 5

$$\begin{aligned} V(t,T) &= \exp\left\{-\int_t^T \delta(s)ds\right\} \\ &= \exp\left\{-\int_t^T \left(0.1 - 0.05e^{-0.1s}\right)ds\right\} \\ &= \exp\left\{-0.1(T-t) - 0.5\left(e^{-0.1T} - e^{-0.1t}\right)\right\} \end{aligned}$$

Hence $PV(2) = 10\,000\,V(2,8) = 10\,000\,e^{-0.1\times 6 - 0.5(e^{-0.8}-e^{-0.2})} = 10\,000\,e^{-0.415299} = 6\,601.43$.

$$\begin{aligned} PV(2) = 10\,000\,V(2,8) &= 10\,000\,e^{-0.1\times 6 - 0.5(e^{-0.8}-e^{-0.2})} \\ &= 10\,000\,e^{-0.415299} \\ &= 6\,601.43 \end{aligned}$$

Solution 6

1. The present value at time 0 is

$$\int_2^7 500 \times e^{-\int_0^t \delta(u)du} dt + 800 \times e^{-\int_0^{1.5} \delta(u)du}$$

If $t < 2$ then

$$e^{-\int_0^t \delta(u)du} = e^{-\int_0^t 0.05du} = e^{-0.05t}$$

and if $t \geq 2$ then

$$e^{-\int_0^t \delta(u)du} = e^{-\int_0^2 \delta(u)du - \int_2^t \delta(u)du} = e^{-\int_0^2 0.05du - \int_2^t 0.07du} = e^{-0.1-0.07(t-2)}$$

Then the present value at time 0 is

$$\begin{aligned}
&\int_2^7 500 \times e^{-0.1-0.07(t-2)} dt + 800 \times e^{-0.05\times 1.5} \\
=&500e^{0.04} \int_2^7 e^{-0.07t} dt + 800 \times e^{-0.075} \\
=&500e^{0.04} \left[\frac{e^{-0.07t}}{-0.07}\right]_2^7 + 800 \times e^{-0.075} \\
=&1908.638 + 742.195 \\
=&2650.83
\end{aligned}$$

2. The accumulated value at time 10 is

$$\begin{aligned}
PV(0)e^{\int_0^{10} \delta(u)du} = 2650.83\, e^{\int_0^2 \delta(u)du + \int_2^{10} \delta(u)du} =& 2650.83\, e^{\int_0^2 0.05du + \int_2^{10} 0.07du} \\
=& 2650.83\, e^{0.05\times(2-0)+0.07\times(10-2)} \\
=& 2650.83\, e^{0.66} = 5128.81
\end{aligned}$$

Solution 7 (i) $\bar{a}_{\overline{5}|} = \frac{1-v^5}{\delta} = 4.15037$
(ii) $\bar{a}_{\overline{9}|} = \frac{1-v^9}{\delta} = 6.49356$
(iii) ${}_{5|}\bar{a}_{\overline{4}|} = v^5\, \bar{a}_{\overline{4}|} = \frac{v^5-v^9}{\delta} = 2.34319$
(iv) $\bar{s}_{\overline{9}|} = (1+i)^9\, \bar{a}_{\overline{9}|} = 12.9807$
(v) $(I\bar{a})_{\overline{15}|} = \frac{\ddot{a}_{\overline{15}|} - 15v^{15}}{\delta} = 58.6740$
(vi) $(\bar{I}\bar{a})_{\overline{15}|} = \frac{\bar{a}_{\overline{15}|} - 15v^{15}}{\delta} = 54.1682$

Solution 8

1. We use integration-by-parts.

$$\begin{aligned}
(\bar{I}\bar{a})_{\overline{n}|} &= \int_0^n t \exp\{-\delta t\}\, dt \\
&= \int_0^n \frac{1}{\delta} \exp(-\delta t)\, dt + \left[\frac{-t}{\delta} \exp(-\delta t)\right]_0^n \\
&= -\frac{n}{\delta} \exp(-\delta n) - \frac{1}{\delta^2}\{\exp(-\delta n) - 1\} \\
&= -\frac{n}{\delta} v^n - \frac{1}{\delta^2} v^n + \frac{1}{\delta^2} \\
&= \frac{1}{\delta}(\bar{a}_{\overline{n}|} - nv^n).
\end{aligned}$$

2. $(\bar{I}\bar{a})_{\overline{n}|}$ is the present value of a continuous cashflow, where the payment rate is t p.a. at time t.

   $(I\bar{a})_{\overline{n}|}$ is the present value of a cashflow where the payment rate is k p.a. between time $k-1$ years and time k years.

   The payment rate in the first case is always smaller than the payment rate in the second case, thus the present value in the first case is smaller than the present value in the second case.

Solution 9

The present value at time 0 is

$$\begin{aligned}
\int_0^{15} \rho(t)e^{-\int_0^t \delta(u)du}dt &= \int_0^{10} \rho(t)e^{-\int_0^t \delta(u)du}dt + \int_{10}^{15} \rho(t)e^{-\int_0^t \delta(u)du}dt \\
&= \int_0^{10} 1000e^{-\int_0^t 0.04du}dt + \int_{10}^{15} 1500e^{-\int_0^t 0.04du}dt \\
&= 1000\int_0^{10} e^{-0.04t}dt + 1500\int_{10}^{15} e^{-0.04t}dt \\
&= 1000\bar{a}_{\overline{10}|} + 1500v^{10}\bar{a}_{\overline{5}|} \\
&= 8242.00 + 4556.57 \\
&= 12798.57
\end{aligned}$$

Alternative 1: $PV(0) = 1500\bar{a}_{\overline{15}|} - 500\bar{a}_{\overline{10}|}$

Alternative 2: $PV(0) = 1000\bar{a}_{\overline{15}|} + 500v^{10}\,\bar{a}_{\overline{5}|}$

Solution 10 We are given that $\rho(t) = 100 - 5t$ and $\delta(t) = 0.1 - 0.005t$, for $t \in [0, 20]$.

We wish to calculate

$$\begin{aligned}
AV(20) &= \int_0^{20} \rho(t)e^{\int_t^{20} \delta(u)du}dt \\
&= \int_0^{20} (100 - 5t)e^{\int_t^{20} (0.1-0.005u)du}dt
\end{aligned}$$

Set $g(t) = \int_t^{20}(0.1 - 0.005u)du$. Recall the Fundamental Theorem of Calculus (FTC): for constant a,

$$\frac{d}{dt}\int_a^t f(u)du = f(t).$$

Then by the FTC,

$$g'(t) = \frac{d}{dt}\int_t^{20}(0.1-0.005u)du = \frac{d}{dt}\int_{20}^t -(0.1-0.005u)du = -(0.1-0.005t) = -0.001(100-5t).$$

Hence

$$AV(20) = \int_0^{20}(100 - 5t)e^{g(t)}dt = \int_0^{20}(-1000g'(t))e^{g(t)}dt.$$

Recall the following result:

$$\frac{d}{dt}\left(e^{g(t)}\right) = g'(t)e^{g(t)}$$

Hence

$$\begin{aligned} AV(20) &= -1000\left[e^{g(t)}\right]_{t=0}^{20} \\ &= 1000\left(e^{g(0)} - e^{g(20)}\right) \\ &= 1000\left(e^{\int_0^{20}(0.1-0.005u)du} - e^0\right) \\ &= 1000\left(e^{[0.1u-0.0025u^2]_0^{20}} - 1\right) \\ &= 1000\left(e^1 - 1\right) = £1718.28 \end{aligned}$$

Solution 11 This is a CT1 question (Q8, September 2010).

1. For $t \leq 20$,

$$V(0,t) = \exp\left\{-\int_0^t (0.05 + 0.001u)du\right\} = \exp\left\{-\left[0.05u + \frac{0.001}{2}u^2\right]_0^t\right\}$$
$$= e^{-0.05t-0.0005t^2}.$$

   For $t > 20$,

$$V(0,t) = \exp\left\{-\int_0^{20} \delta(u)du - \int_{20}^t 0.05du\right\} = v(20)\ \exp\left\{-\left[0.05u\right]_{20}^t\right\}$$
$$= e^{-1.2}e^{1-0.05t} = e^{-0.2-0.05t}.$$

2. At time $t = 0$,

   (a) $PV = 100V(0,25) = 100e^{-0.2-0.05\times 25} = 100e^{-1.45} = £23.46$.

   (b)

$$100\left(1 - \frac{d^{(4)}}{4}\right)^{4\times 25} = 100V(0,25) = 23.46.$$

   Hence $d^{(4)} = 4(1 - 0.2346^{1/100}) = 0.05758$.

3. Set $\rho(t) = 30e^{-0.15t}$. Then

$$\begin{aligned} AV(25) &= \int_{20}^{25} \rho(t)e^{\int_t^{25}\delta(u)du}dt \\ &= \int_{20}^{25} 30e^{-0.015t}e^{\int_t^{25}0.05du}dt \end{aligned}$$

   Since $\int_t^{25} 0.05du = [0.05u]_t^{25} = 1.25 - 0.05t$, then

$$\begin{aligned} AV(25) &= 30\int_{20}^{25} e^{-0.015t}e^{1.25-0.05t}dt \\ &= 30e^{1.25}\int_{20}^{25} e^{-0.065t}dt \\ &= 30e^{1.25}\left[\frac{e^{-0.065t}}{-0.065}\right]_{20}^{25} \\ &= 30e^{1.25}\left(\frac{e^{-1.625} - e^{-1.3}}{-0.065}\right) \\ &= 121.82 \end{aligned}$$